

REPORT ON SUPERSTORE MARKETING CAMPAIGN FOR GOLD CARD

Part 1 — Executive Summary + KPI Snapshot

Part 2 — Customer Segmentation Findings

Part 3 — Spending & Channel Analysis

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Part 1 — Executive Summary

Superstore Marketing Campaign Analysis Executive Summary

Background

The Superstore chain runs an annual year-end promotional campaign offering existing customers a Gold Membership with a 20% discount on all purchases. This analysis was conducted on a dataset of 2,213 customers across 22 features to understand what drives membership acceptance, identify high-value segments, and recommend targeted marketing strategies to improve campaign ROI.

Key KPIs at a Glance

KPI	Value
Total Customers Analysed	2,213
Overall Membership Acceptance Rate	15.04%
Average Customer Income	₹52,247
Average Total Spend (2 years)	₹606.78
Average Recency	49.1 days
Top Spending Category	Wine (avg ₹303)
Most Used Purchase Channel	In-Store (39%)
Complaint Rate	1.06%

What This Analysis Covers

This report presents findings across four areas:

- Customer segmentation by demographics and behavior
- Spending patterns across product categories and channels
- Predictive modelling for membership acceptance
- Strategic recommendations for the next campaign

Part 2 — Customer Segmentation Findings

Customer Demographics Overview

The customer base skews older with a mean age of 57 years (range 30–86), is predominantly married or in a relationship (65%), and is highly educated — over 93% hold at least a graduation degree. Most households are small, with a median family size of 1 (kids + teens combined).

Segmentation by Education

Education	Count	Share
Graduation	1,116	50.4%
Master	564	25.5%
PhD	480	21.7%
Basic	54	2.4%

Chi-square test confirmed education significantly associates with membership acceptance ($\chi^2=21.33$, $p=0.0001$). Higher education levels show greater tendency to accept the Gold Membership, with PhD and Master holders being the most receptive segments.

Segmentation by Marital Status

Marital Status	Count	Share
Married	857	38.7%
Together	572	25.8%
Single	477	21.5%
Divorced	232	10.5%
Widow	76	3.4%

Chi-square test confirmed marital status significantly associates with response ($\chi^2=51.14$, $p=0.0000$). Single and divorced customers show relatively higher acceptance rates compared to married customers with children.

Segmentation by Income

T-test confirmed income significantly separates responders from non-responders (t=6.32, p=0.0000):

Segment	Average Income
Responders	₹60,210
Non-Responders	₹50,824
Difference	+₹9,386 (18.5% higher)

Higher income customers are substantially more likely to accept the membership — income is one of the top 4 predictors in the model.

Segmentation by Family Composition

Family size emerged as the **6th most important feature** in the SMOTE model, jumping significantly from the original RF. Key finding — customers with no children or teenagers show higher acceptance rates. Households with kids tend to prioritize differently and are less receptive to the Gold Membership offer.

Customer Profiles Identified

Based on the segmentation analysis, four distinct profiles emerge:

-  **High-Value Loyalists** — High income (>₹60k), high total spend, low recency, educated. Most likely to accept membership. Primary target segment.
-  **Engaged Regulars** — Medium income, moderate spend, frequent store and web purchases. Receptive with the right incentive.
-  **Bargain Hunters** — High deal purchases, low catalogue usage, price-sensitive. Less likely to accept unless membership is framed as a savings tool.
-  **Dormant Customers** — High recency (>60 days), low spend. re-engagement before membership pitch.

Part 3 — Spending & Channel Analysis

How Much Are Customers Spending?

Total spend across all product categories over the last two years tells us a lot about customer value and membership likelihood.

Metric	Value
Average Total Spend	₹606.78
Median Total Spend	₹396.50
Minimum Total Spend	₹5.00
Maximum Total Spend	₹2,525.00

What this means that: The gap between mean (₹606) and median (₹396) tells us the distribution is right-skewed — a small group of high spenders is pulling the average up. Most customers actually spend below ₹400. This means a relatively small high-value segment is contributing disproportionately to revenue — protecting and growing this segment should be a priority.

Which Product Categories Drive Spend?

Category	Avg Spend (All Customers)	Avg Spend (Responders)	Avg Spend (Non-Responders)
Wine	₹303.94	₹534.27	₹262.44
Meat Products	₹166.95	₹311.20	₹140.61
Gold Products	₹44.02	₹82.45	₹36.88
Fish Products	₹37.63	₹72.31	₹30.67
Sweet Products	₹27.06	₹48.91	₹22.58
Fruits	₹26.36	₹48.20	₹21.58

What this means that: Responders spend more across every single category — but the gap is most dramatic in Wine and Meat. A customer who spends heavily on wine and meat is sending a very strong signal that they value quality and are willing to pay for it. These are exactly the customers who see the most value in a 20% across-the-board discount.

Takeaway: Wine and meat spending are strong behavioral indicators of membership acceptance. Customers spending above average on these two categories should be prioritised in targeting.

Category Breadth — Are Customers Buying Across Categories?

Category breadth measures how many of the 6 product categories a customer spends on.

Category Breadth	Meaning
Median: 6 out of 6	Most customers buy across ALL categories
Mean: 5.43	Very high cross-category engagement
Minimum: 2	A small group buys from only 2 categories

What this means that: The majority of Superstore customers are not single-category shoppers — they buy wine, meat, fish, fruits, sweets and gold products. This is actually great news for cross-selling. These customers already have broad product affinity — the membership just needs to be positioned as a way to save money on purchases they're already making.

Which Channels Are Customers Using?

Channel	Share of Total Purchases	Avg (Non-Responders)	Avg (Responders)
In-Store	39.0%	5.76	6.08
Web	27.5%	3.91	5.07
Catalogue	17.9%	2.40	4.20
Deals/Discounts	15.6%	2.32	2.34

What this means that:

In-Store is the dominant channel overall but doesn't strongly separate responders from non-responders. Everyone shops in-store — it's not a differentiator for targeting.

Web purchases show a meaningful gap — responders average 5.07 web purchases vs 3.91 for non-responders. Digitally engaged customers are more receptive. This makes sense — they're already comfortable with online interaction and likely more exposed to digital campaign touchpoints.

Catalogue is the biggest differentiator — responders make 4.20 catalogue purchases on average vs only 2.40 for non-responders. That's a 75% higher usage

rate. Catalogue customers tend to be deliberate, high-intent buyers who browse and plan their purchases — exactly the profile that values a membership discount.

Deals and web visits show virtually no difference between responders and non-responders. Discount hunters and casual browsers are not your membership audience — targeting them heavily wastes marketing budget.

Channel Strategy Summary

Channel	Action
Catalogue	Primary campaign channel — highest differentiator
Web	Strong secondary channel — target digitally active customers
In-Store	Use for awareness and point-of-sale prompts, not primary targeting
Deals/Discounts	Deprioritise for membership campaign — wrong audience

Spending + Channel Combined Insight

The most valuable customer profile emerging from this analysis is someone who spends heavily on wine and meat, makes frequent catalogue and web purchases, and has shopped recently. This customer is not price-sensitive in the traditional sense — they spend freely but respond to perceived value. A 20% Gold Membership discount is not a coupon to them — it's a smart financial decision on purchases they were going to make anyway.

Part 4 — Predictive Model Summary & Comparison

Why Build a Predictive Model?

Statistical tests tell us *which factors* are associated with membership acceptance. A predictive model goes further — it combines all those factors simultaneously to answer: *"Given everything we know about this customer, how likely are they to accept?"* This allows the marketing team to score and rank every customer before the campaign launches.

The Challenge — Class Imbalance

The dataset contains 1,880 non-responders vs only 333 responders (85:15 ratio). This means a model can achieve 85% accuracy by simply predicting "No" for everyone — which is useless for marketing. We addressed this using **SMOTE** (Synthetic Minority Oversampling Technique), which artificially balances the training data so the model learns to identify responders more effectively.

Model Comparison

Metric	Logistic Regression	Random Forest	RF + SMOTE 
Accuracy	86%	86%	84%
Precision (Responders)	63%	65%	47%
Recall (Responders)	18%	22%	48%
F1 Score (Responders)	0.28	0.33	0.47
ROC-AUC	0.80	0.84	0.83
Responders Correctly Caught	~12	15	32

Why RF + SMOTE is the recommended model: It catches 32 out of 67 actual responders in the test set — more than double the original Random Forest. Yes, accuracy drops slightly from 86% to 84%, but in a marketing context missing a genuine prospect is far more costly than sending an extra offer to someone who declines.

What Does the Model Say Drives Acceptance?

These are the top features ranked by importance in the final model:

Rank Feature		Business Meaning
1	Recency	Recent shoppers are far more likely to accept
2	Web/Store Ratio	Digitally engaged customers respond better
3	Total Spend	Higher spenders see more value in the discount
4	Income	Higher income = more comfortable with membership cost
5	Tenure Days	Longer-standing customers are more loyal and receptive
6	Family Size	Smaller households respond better
7	Store Purchases	Frequent in-store shoppers are engaged customers

Model Limitations — Being Honest

- The model still misses 35 out of 67 responders (52%) — it's a starting point, not a perfect filter
 - Dataset size (2,213 records) is relatively small for machine learning
 - Richer features like browsing history, past campaign responses, or transaction timestamps would significantly improve performance
 - For production use, the model should be retrained periodically as new campaign data becomes available
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Part 5 — Strategic Recommendations

Overview

The analysis consistently points to one central truth — the Gold Membership offer is not for everyone, and that's okay. The goal is not to convince every customer but to identify and reach the right ones efficiently. The following recommendations are directly backed by the data.

Recommendation 1 — Target High-Income, High-Spend Customers First

The data is unambiguous — responders earn ₹9,386 more annually and spend nearly twice as much as non-responders. Build a priority tier of customers with income above ₹55,000 and total spend above ₹400. This segment has the highest conversion probability and should receive the most personalised, high-effort outreach.

Expected impact: Higher conversion rate, better campaign ROI, lower cost per acquisition.

Recommendation 2 — Lead With Catalogue & Web Channels

Catalogue customers are 75% more likely to be responders. Web-active customers also show significantly higher acceptance. These are deliberate, high-intent buyers who plan their purchases — exactly the audience for a value-based membership offer.

Action: Prioritise catalogue mailouts and targeted web campaigns over blanket in-store promotions. Reduce budget allocation toward deal/discount channels — the data shows they attract the wrong audience for this offer.

Recommendation 3 — Act on Recency — Time the Campaign Right

Recency is the single most important predictor in the model. Customers who shopped within the last 30-35 days are significantly more likely to accept. They are actively engaged and top of mind with the brand.

Action: Trigger membership offers within 2 weeks of a customer's last purchase while brand engagement is at its peak. Avoid reaching out to customers who haven't shopped in 60+ days without a re-engagement campaign first.

Recommendation 4 — Re-engage Dormant Customers Separately

Dormant customers (recency > 60 days, low total spend) are unlikely to respond to a membership pitch. Pushing the Gold Membership to them prematurely wastes budget and risks further disengagement.

Action: Run a separate re-engagement campaign first — a simple discount offer or product recommendation based on their past category spend. Once they make a purchase, they enter the high-recency window and become viable membership targets.

Recommendation 5 — Rethink the Offer for Family Households

Households with children and teenagers show lower acceptance rates. The current membership benefits (discounts on wine, meat, gourmet products) don't naturally align with family spending priorities.

Action: Either deprioritise family households in the main campaign, or create a parallel family-oriented membership tier with benefits relevant to their actual purchases — everyday essentials, fruits, sweets. Don't push the same offer to fundamentally different customer needs.

Recommendation 6 — Use the Predictive Model to Score & Rank Customers

Rather than sending the campaign to all 2,213 customers, use the RF + SMOTE model to score every customer by their probability of accepting. Focus the high-effort personalised outreach on the top 20-30% of scores.

Action: This approach reduces campaign costs while concentrating effort where conversion is most likely. As new campaign data comes in, retrain the model to continuously improve targeting accuracy.

Summary of Recommendations

Priority Recommendation		Data Backing
1	Target income >₹55k, spend >₹400	T-test, Feature Importance
2	Lead with Catalogue & Web channels	Channel Analysis
3	Time offers within 30 days of last purchase	Recency — #1 predictor

Priority Recommendation

4 Re-engage dormant customers separately

5 Redesign offer for family households

6 Score customers using predictive model

Data Backing

Recency & Spend Analysis

Family Size — Feature Importance

RF + SMOTE, ROC-AUC 0.83
